

MATH 15200: Calculus II

Homework 1: Limits and Continuity

Due: Friday, October 9, 2026, at 11:59 PM.

This is the written portion of Homework 1; complete the routine online portion separately in Edfinity. Upload your written solutions to Gradescope, via Canvas. You may collaborate, but you must write and submit your own solutions independently.

Definition. Let f be a real-valued function defined near a , possibly not at a . We say

$$\lim_{x \rightarrow a} f(x) = L$$

if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that, whenever x is in the domain of f ,

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

If $f(a)$ is defined, then f is **continuous at** a when

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Rewrite each of the problems below in terms of the epsilon-delta definition above. Explicitly state your choice of δ when necessary.

1. Prove directly from the epsilon-delta definition that

$$\lim_{x \rightarrow 2} (x^2 - 1) = 3.$$

Hint: Impose a simple bound such as $|x - 2| < 1$.

2. Define a function f by the piecewise expression

$$f(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Prove directly from the definition that f is not continuous at 0.

Hint: Choose a fixed positive ε and consider points immediately to the left of 0.

3. Suppose that f and g are real-valued functions defined near a and that

$$\lim_{x \rightarrow a} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = M.$$

Prove from the epsilon-delta definition that

$$\lim_{x \rightarrow a} (f(x) + g(x)) = L + M.$$

Hint: Apply the two hypotheses using $\varepsilon/2$ and combine the resulting choices of δ .